

Three Experiments for the Universidad de Cuenca

A concept note on rigorous, ethical, university-led replication studies of consciousness and altered states — proposed in alignment with the University's mission and vision.

Institution	Universidad de Cuenca · Cuenca, Ecuador
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Purpose	Rough proposals and indicative budgets to support early conversations with university partners.
Status	Draft for discussion — figures are order-of-magnitude estimates.

Framing. All three studies are designed as *replication / falsification* studies. The aim is not to prove extraordinary claims, but to test them under conditions that are publishable even when the result is negative. A clean null result is a valuable scientific contribution; a positive result would be highly significant.

Costing basis. Budgets are grounded in **Cuenca, Ecuador** conditions: local academic and clinical salaries, and the assumption that major equipment (EEG, fMRI, infusion pumps, monitoring) is **borrowed from partner institutions rather than purchased**. Equipment lines therefore cover access, transport, calibration, and consumables only. All figures are in USD (Ecuador's currency). Each study is presented as a range — a **partnership-leveraged** lower bound and a **fully-costed** upper bound (explained on the comparative budget page).

Alignment with the Universidad de Cuenca

This research agenda is proposed specifically for the **Universidad de Cuenca**. It is designed not merely to sit alongside the University's mission and vision, but to express them. We are on the same path.

Mission · Universidad de Cuenca

"To train researchers and professionals committed to a just, diverse and sustainable society, ready to be agents of transformation."

Vision 2027

"By 2027 the Universidad de Cuenca is an innovative and resilient university community, integrated with the world through the generation of relevant, high-quality knowledge committed to society."

Official mission and vision, Universidad de Cuenca (ucuenca.edu.ec).

How this programme advances both

Generation of relevant, high-quality knowledge. Preregistered, publishable studies at the frontier of neuroscience produce exactly the kind of original, socially meaningful knowledge the vision calls for.

Integrated with the world. Open-data, internationally collaborative designs connect Cuenca to the global community studying consciousness, inviting partnerships across laboratories and disciplines.

Agents of transformation. Training graduate researchers inside a bold yet ethically grounded programme forms precisely the transformative scientists the mission describes.

Innovative and resilient. Testing extraordinary claims under clean, falsifiable conditions is innovation done responsibly — and resilient, because even a negative result is a genuine contribution.

Global Impact — Why Studies of This Kind Matter

Consciousness remains one of the last great frontiers of science. Studying it rigorously — through extended psychedelic states, brain-to-brain paradigms, and the neurophysiology of trance — speaks to fundamental questions about mind, brain, and human experience, with consequences that reach well beyond any single laboratory.

Mental health. Extended-state and psychedelic research is already reshaping treatments for depression, PTSD, addiction, and end-of-life distress; rigorous local data contributes directly to that global effort.

A model for how science meets extraordinary claims. Bringing controversial-but-important questions into the laboratory under preregistration and open data shows how science should test bold claims — neither dismissing nor inflating them. That methodological example is itself a contribution to scientific culture.

Knowledge at the frontier. A positive result would be a landmark finding; a well-powered null result clarifies the limits of these phenomena. Either way the knowledge is new, citable, and field-shaping.

Cuenca on the global map. A university in Ecuador leading careful, ethical consciousness research is a statement that frontier science belongs everywhere — drawing collaboration, students, and international visibility to the Universidad de Cuenca.

Executive Summary

This note outlines three candidate experiments that could anchor a research agenda at the Universidad de Cuenca on altered states of consciousness, neurophenomenology, and the scientific study of extraordinary human experience. Each is summarised below with a core idea, a proposed design, required infrastructure, ethical considerations, and an indicative budget. A comparative budget table and a longer pipeline of ten further candidates are included at the end.

Study	Core question	Indicative budget	Risk / complexity
1. Extended-State DMT (DMTx)	Can the DMT state be safely extended and characterised in real time?	\$40k – \$150k	High — controlled substance, medical
2. Brain-to-Brain Correlation	Do separated, paired subjects show correlated EEG responses?	\$17k – \$55k	Medium — design rigor is everything
3. Trance / Healer Neurophysiology	Do practitioners enter a reproducible, distinct brain state?	\$18k – \$50k	Low–Medium — recruitment-limited

Recommended sequencing. Studies 2 and 3 are lower-cost, lower-risk, and faster to ethics approval; either could serve as a credible first project that demonstrates rigor to partners and funders. Study 1 (DMTx) is the scientific flagship but carries the heaviest regulatory, medical, and budgetary load and is best pursued once institutional relationships and a clinical partner are in place.

Extended-State DMT / DMTx

Core idea

Rather than studying DMT as a very short (5–15 minute) experience, this protocol uses a controlled intravenous loading dose followed by a target-controlled maintenance infusion to **extend and stabilise the DMT state** for an experimentally useful window. This makes it possible to study the phenomenology, neural correlates, and any recurring structures of the experience while collecting real-time physiological and neuroimaging data. The informal DMT / laser-perception idea we discussed would sit inside this framework as a small, blinded perceptual sub-study rather than as the flagship aim.

Proposed design

- **Design:** open-label pharmacokinetic / pharmacodynamic pilot, $n \approx 8$ –12 healthy, psychedelic-experienced volunteers, within-subject dose escalation.
- **Primary outcomes:** safety and tolerability of extended infusion; feasibility of maintaining a stable subjective state at a target plasma level.
- **Secondary outcomes:** EEG signatures (spectral power, complexity / entropy, connectivity); validated phenomenology scales; structured post-session interviews.
- **Optional blinded sub-study:** randomised, blinded perceptual tasks delivered during the stable state to test specific claims under controlled conditions.
- **Analysis:** preregistered, with predefined safety stopping rules and primary endpoints.

Key infrastructure & equipment

- Clinical research unit with anaesthesia-grade monitoring (partner hospital)
- Target-controlled IV infusion pump(s) — borrowed
- Research-grade EEG & physiological monitoring — borrowed
- Pharmaceutical-grade DMT and hospital pharmacy handling
- Controlled-substance storage and chain-of-custody

Ethical & safety considerations

- Controlled-substance licensing and import permits (Ecuadorian regulation)
- Full research-ethics board (IRB) review
- On-site physician and resuscitation capability
- Rigorous screening; cardiac and psychiatric exclusions
- Trained psychological support and integration

Indicative timeline

Approximately 18–24 months: 6–9 months for regulatory, ethics, and drug supply; 3 months setup and staff training; 6–9 months data collection; 3–6 months analysis and write-up.

Indicative budget (rough / general)

Cost category	Estimate (USD)
Regulatory, ethics & controlled-substance permits (Ecuador)	\$8,000
Pharmaceutical-grade DMT supply & secure handling	\$20,000

Clinical & medical staff (physician, anaesthesia, nurse, psychiatrist) — local rates	\$25,000
Borrowed EEG / monitoring — access, transport, calibration & consumables	\$8,000
Core research personnel (PI, postdoc, RAs, analyst — ~1 yr, Cuenca rates)	\$40,000
Participant screening, compensation & facility	\$8,000
Data analysis, computing, dissemination & contingency	\$6,000
Indicative total	≈ \$90,000 – \$150,000 (fully costed)

With maximal in-kind institutional support — donated drug synthesis, clinical collaborators contributing as co-investigators, borrowed instrumentation, and graduate researchers embedding the work within their theses — the partnership-leveraged floor falls to approximately \$40,000.

Key references & links

- Gallimore & Strassman (2016), [Frontiers in Pharmacology](#)
- Extended DMT study ([PMC10851633](#))
- Continuous DMT infusion study ([PMC12032411](#))

Brain-to-Brain Correlation / “Transferred Potential” Replication

Core idea

Inspired by Jacobo Grinberg’s controversial work on “transferred potentials,” in which two people are placed in separate rooms, one receives a stimulus, and the other’s brain activity is monitored for correlated responses. The claim is extraordinary — which is exactly why it deserves a **modern, rigorously controlled replication**. The scientific value here lies almost entirely in design quality: double-blinding, shielding, randomised timing, and fully automated, preregistered analysis remove the loopholes that made the original work impossible to interpret.

Proposed design

- **Design:** paired-subject EEG study, $\approx$ 25–30 pairs (emotionally bonded and stranger pairs), “sender” and “receiver” in separate electromagnetically shielded rooms.
- **Stimulus:** randomised, computer-triggered sensory stimuli to the sender at times unknown to the receiver and to all experimenters present.
- **Primary outcome:** preregistered test for time-locked correlated activity in the receiver’s EEG during sender-stimulation vs. matched null windows.
- **Controls:** sham/no-sender blocks, sensor-level artifact rejection, and analyst blinding to condition labels.
- **Statistics:** preregistered effect-size threshold, Bayesian + frequentist criteria; a well-powered null is a publishable outcome.

Key infrastructure & equipment

- Two research-grade EEG systems, time-synchronised — borrowed
- Electromagnetically shielded / Faraday-isolated rooms (existing facility)
- Hardware trigger & precision timing infrastructure
- Automated, preregistered analysis pipeline
- Audio/RF isolation verification

Ethical & safety considerations

- IRB / ethics approval (low physical risk)
- Informed consent; right to withdraw
- Pre-registration on OSF / AsPredicted before data collection
- Independent verification of room isolation
- Open data and analysis code for full transparency

Indicative timeline

Approximately 12–15 months: 2–3 months ethics and preregistration; 2 months setup and isolation testing; 5–6 months paired data collection; 3–4 months analysis and write-up.

Indicative budget (rough / general)

Cost category	Estimate (USD)
Borrowed EEG — transport, caps/electrodes, gel, calibration	\$4,000
Shielded-room setup & isolation testing (existing facility)	\$5,000

Core personnel (PI part-time, postdoc, 2 RAs, analyst — ~1 yr, Cuenca rates)	\$30,000
Participant recruitment & compensation (~60 participants)	\$4,000
Preregistration, replication design & statistical consulting	\$3,000
Computing, open-data hosting & dissemination	\$2,000
Indicative total	≈ \$35,000 – \$55,000 (fully costed)

With borrowed instrumentation, free use of an existing shielded facility, and student researchers carrying the data collection, the partnership-leveraged floor falls to approximately \$17,000.

Key references & links

- [Jacobo Grinberg — background & publication reference \(Wikipedia, ES\)](#)
- [Related separated-subject EEG study \(PubMed 12972348\)](#)

Shamans / Healers / Trance Practitioners under EEG or fMRI

Core idea

Study shamans, healers, meditators, or trance practitioners while they enter their claimed healing or altered state, and compare their brain activity and physiology against matched controls and against their own ordinary rest, imagination, and meditation. Crucially, this study does **not** need to begin by claiming that “healing energy” is real. The first, stronger scientific question is simply: *do trained practitioners enter a reproducible neurophysiological state that is measurably different from ordinary baselines?* If yes, a later phase can test whether that state has any measurable effect on recipients.

Proposed design

- **Design:** within-subject, multi-condition comparison (trance vs. rest vs. imagery vs. meditation), ≈ 15–25 experienced practitioners + matched controls.
- **Primary outcome:** reproducible, practitioner-specific EEG/fMRI signature that discriminates the trance state from control conditions above chance.
- **Secondary outcomes:** autonomic measures (HRV, electrodermal activity, respiration); state classification with cross-validated machine learning.
- **Phase 2 (optional):** blinded recipient study testing for any physiological or perceptual effect on a second person.
- **Analysis:** preregistered classification accuracy and effect-size thresholds.

Key infrastructure & equipment

- Research-grade EEG and/or fMRI scanner access — borrowed
- Autonomic / peripheral physiology recording (HRV, EDA) — borrowed
- Synchronised audio-video for state annotation
- Analysis workstation & ML pipeline
- Quiet, controlled recording environment

Ethical & safety considerations

- IRB / ethics approval; culturally respectful protocol
- Community consultation & fair compensation for practitioners
- Informed consent honouring practitioners' traditions
- Clear non-medical-claim framing in all materials
- Pre-registration of hypotheses and analysis

Indicative timeline

Approximately 12–18 months: 2–3 months ethics, community engagement, and recruitment; 5–7 months data collection (recruitment-paced); 3–5 months analysis and write-up.

Indicative budget (rough / general)

Cost category	Estimate (USD)
Borrowed EEG / fMRI — scanner access fees & consumables (~40 sessions)	\$5,000
Core personnel (postdoc, RAs, data analyst — ~1 yr, Cuenca rates)	\$28,000

Practitioner recruitment, travel & compensation	\$6,000
Peripheral physiology (HRV, EDA) — borrowed / low-cost	\$2,000
Ethics, preregistration, dissemination & contingency	\$3,000
Indicative total	≈ \$30,000 – \$50,000 (fully costed)

With donated scanner / EEG time, borrowed physiology hardware, and student researchers leading data collection, the partnership-leveraged floor falls to approximately \$18,000.

Key references & links

- [Cognitive / shamanic-style trance research \(arXiv 2509.19254\)](#)
- [Trance & neurophysiology — general background \(Wikipedia\)](#)

Comparative Budget Overview

The figures below are **rough, order-of-magnitude estimates** for pilot studies run in **Cuenca, Ecuador**, using **local salaries** and **borrowed equipment** from partner institutions (no major hardware purchases). Each study is shown across two scenarios: a **partnership-leveraged** figure that assumes maximal in-kind support, and a **fully-costed** figure at market rate. The largest variable in both is skilled personnel time.

Study	Partnership-leveraged	Fully costed	Duration	Role
1 • Extended-State DMT (DMTx)	~\$40k	\$150k	18–24 months	Flagship — later phase
2 • Brain-to-Brain Correlation	~\$17k	\$55k	12–15 months	Strong candidate
3 • Trance / Healer Neuropsychology	~\$18k	\$50k	12–18 months	Strong candidate
Combined programme (all three)	~\$75k	\$255k	~2–3 years phased	—

The *partnership-leveraged* column reflects maximal in-kind support (donated equipment and scanner time, collaborators contributing as co-investigators, student-led data collection); the *fully-costed* column prices every line at market rate. A realistic operating budget usually lands between the two, since skilled personnel time is the least compressible cost. Studies 2 and 3 are modest enough to run as graduate / faculty research projects; Study 1 (DMTx) carries the clinical, drug, and regulatory load that keeps it the most expensive even at local prices.

Cross-Cutting Principles

Replication, not advocacy. Every study is framed to be publishable whether the result is positive or null. We are testing claims, not defending them.

Preregistration by default. Hypotheses, sample sizes, and analysis plans are registered before data collection on OSF or a comparable platform, protecting against bias and post-hoc storytelling.

Ethics first. Each study is designed to be ethically approvable at a mainstream university, with appropriate review, consent, safety, and (for Study 3) cultural respect.

Open and transparent. Open data and analysis code wherever ethically possible, so results can be independently verified — the single most important safeguard for controversial questions.

Broader Pipeline — Ten Further Candidates

Beyond the three priority studies, the following are strong additional replication candidates in consciousness and altered-states research, available to build out a longer programme:

1. Real-time communication with lucid dreamers during REM sleep
2. Psilocybin and increased brain-state repertoire
3. Ayahuasca and brain entropy / network reorganization
4. Expert meditators and high-amplitude gamma synchrony
5. Hypnosis and changes in pain, agency, and self-perception
6. Breathwork-induced non-ordinary states without psychedelics
7. Ganzfeld / sensory deprivation and internally generated perception
8. Near-death experience research and end-of-life brain activity
9. Anesthesia and perturbational complexity as a measure of consciousness
10. Virtual-reality body-transfer or out-of-body illusion experiments

Prepared by Ney Torres · 23 June 2026 · Draft concept note for discussion with the Universidad de Cuenca. All budget figures are indicative and subject to refinement with the host institution.